Dan Ley

Interpretable language models • Training data attribution • Faithful reasoning

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Research experience

Guide Labs Research Scientist Intern 2025–2026

- **Prototype Language Models (PRISM).** Designed a new language model architecture that connects predictions to training examples in a single forward pass. Led implementation, training, evaluation (COLM 2026). Supervised by Julius Adebayo.
- **Scaling and evaluation.** Trained PRISM with up to 1.6B parameters. After 10B FineWeb tokens, the largest model matched the black-box baseline on average benchmark accuracy (48.95 versus 48.89) while retaining 97.3% of its training throughput.
- **Training data attribution.** Developed a prototype space method to identify influential training examples, with cached scoring 470× faster than EK-FAC in a controlled 124M-model comparison under the same GPU memory budget. Analyzed how clustering losses regularize the prototype space Hessian, improving conditioning and block-diagonal approximation accuracy.

Harvard University Doctoral Researcher Sep 2022–present

- **Grouped Gradient Influence (GGInf).** Developed a method to reduce the cost of training data attribution by letting similar training examples share stored representations, rather than retaining one per example (reducing storage/serving from $O(Nd)$ to $O(Kd)$).
- **Theory and experiments.** Derived spectral bounds on attribution error and evaluated grouping with LoGRA, empirical Fisher influence, and TRAK. Preserved attribution fidelity at up to 40× additional storage reduction beyond existing parameter compression on classification benchmarks, and 32× on Pythia-1B with 3 million candidate examples (experiments up to Pythia-6.9B).
- **Faithful reasoning.** Co-led research testing whether language models' stated reasoning reflects how they reach answers. Evaluated prompting, finetuning, and activation editing; showed how finetuning complicates this goal by changing the underlying model behavior being explained. Cited in OpenAI's o1 System Card.
- **Data attribution and tools.** Summarized mathematical formulations and computational cost comparisons for major attribution methods in a survey accepted to JAIR (2026). Repaired the OpenXAI evaluation codebase, corrected erroneous experimental results, and coauthored the revised arXiv paper (prior to agentic coding).

JPMorgan Chase & Co. AI Research Intern Oct 2021–Jul 2022

- **GLOBE-CE.** Designed a method for global counterfactual explanations, identifying changes that reverse model predictions across many individuals (for example, changes in income or debt that alter loan default predictions). Developed the theory for categorical features and led implementation, experiments, and the first-author ICML 2023 paper with Saumitra Mishra and Daniele Magazzeni.
- Six-month internship extended to nine months; received a full-time research offer before choosing to pursue the Harvard PhD.

Earlier research and engineering

University of Cambridge Research Assistant Jul–Sep 2021

Developed diverse and amortised counterfactual explanations of predictive uncertainty (first-author AAAI 2022 paper).

JPMorgan Chase & Co. Software Engineering Intern Jul–Aug 2020

Python software development with Flask, SQL, and automated testing.

Imagination Technologies Hardware Engineering Intern Jul–Sep 2019

Co-inventor on 3 patents for faster, more compact arithmetic hardware, including floating point squaring and efficient squaring circuits.

Education

Harvard University PhD, Computer Science 2022–2027 expected

Advisor: Himabindu Lakkaraju

University of Cambridge MEng, Engineering 2017–2021

Distinction; 12th of 324 in Year 1 and 12th of 310 in Year 2 (top 4% both years).
Outstanding Research Project Award (top 5%) in Information Engineering. Advisors: Adrian Weller and Umang Bhatt.

Publications

Selected publications and manuscripts

Prototype Language Models

COLM 2026

Dan Ley, Giang Nguyen, Himabindu Lakkaraju, Julius Adebayo

Grouped Gradient Influence: Compressing the Data Axis of Training Data Attribution

Manuscript, 2026

Dan Ley, Shichang Zhang, Himabindu Lakkaraju

A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI

JAIR, accepted 2026

Junwei Deng et al. (including **Dan Ley**)

On the Hardness of Faithful Chain-of-Thought Reasoning in Large Language Models

2024 preprint

Sree Harsha Tanneru*, **Dan Ley***, Chirag Agarwal, Himabindu Lakkaraju

On Minimizing the Impact of Dataset Shifts on Actionable Explanations

UAI 2023 (oral, top 5%)

Anna P. Meyer*, **Dan Ley***, Suraj Srinivas, Himabindu Lakkaraju

GLOBE-CE: A Translation Based Approach for Global Counterfactual Explanations

ICML 2023

Dan Ley, Saumitra Mishra, Daniele Magazzeni

Diverse, Global and Amortised Counterfactual Explanations for Uncertainty Estimates

AAAI 2022

Dan Ley, Umang Bhatt, Adrian Weller

OpenXAI: Towards a Transparent Evaluation of Model Explanations

NeurIPS D&B 2022

Chirag Agarwal et al. (including **Dan Ley**)

Other selected contributions

In-Context Explainers: Harnessing LLMs for Explaining Black Box Models

2024 preprint

Nicholas Kroeger*, **Dan Ley***, Satyapriya Krishna, Chirag Agarwal, Himabindu Lakkaraju

Consistent Explanations in the Face of Model Indeterminacy via Ensembling

ICML workshop 2023

Dan Ley, Leonard Tang, Matthew Nazari, Hongjin Lin, Suraj Srinivas, Himabindu Lakkaraju

Degraded Polygons Raise Fundamental Questions of Neural Network Perception

NeurIPS D&B 2023

Leonard Tang, **Dan Ley**

δ-CLUE: Diverse Sets of Explanations for Uncertainty Estimates

ICLR workshop 2021

Dan Ley, Umang Bhatt, Adrian Weller

* Equal contribution. Full publication record: Google Scholar link on page 1.

Technical background and distinctions

Technical. Python, PyTorch, LLM training, influence functions and curvature/gradient-based attribution, counterfactual explanations.

Honours. NeurIPS 2023 Top Reviewer (approx. top 10%); Corpus Christi College Scholar; Dewhurst Scholar (2018–2021).

Mathematics. 90% average in highest Cambridge mathematics modules (Years 1–3); British Mathematical Olympiad (top 500 UK); Senior Team Mathematics Challenge national finalist (2016, 2017).

Sport. Cambridge University Blues football (2017–2021); MIT FC, Bay State Division 1 (2022–2024); Corpus Christi FC coach (2021–2022); marathon and double-marathon runner.

Games and puzzles. Lichess peak rating: 2025 elo (bullet); Expert Killer Sudoku 3:26; Call of Duty Mobile Global Top 100 (Man-O-War leaderboard, 2021); JobFlare: 1st of 220,000 users (cognitive speed tests, 2019); 50,000 Brilliant.org interactions on mathematics problems/solutions (2017).